Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 33)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

0 1 . 2 Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

0 1 . 3 Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

0 1 . 4 Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

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0 2 . 1 Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

0 2 . 2 Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

0 2 . 3 Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

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0 3 . 1 Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

Square-mesh escape panels

Minimum and maximum catchable sizes

No-take zones

Closed seasons

0 3 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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0 4 . 1 State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

0 4 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

0 4 . 3 Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

0 4 . 4 Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

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0 5 . 1 Explain how each of the following agricultural changes affects wildlife habitats.

[6 marks]

Drainage of wet grassland

Application of fertiliser to a species-rich meadow

Removal of hedgerows to enlarge fields

0 5 . 2 Explain why some farmland habitats are valuable for wildlife conservation.

[2 marks]

0 5 . 3 Describe how agriculture contributes to climate change.

[2 marks]

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0 6 . 1 Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

0 6 . 2 Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

0 6 . 3 Explain **three** ways in which coral reefs are important to humans.

[3 marks]

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0 7 . 1 Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

Light .....

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Dissolved oxygen .....

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Nutrients .....

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0 7 . 2 Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

[4 marks]

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0 8 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 8 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 8 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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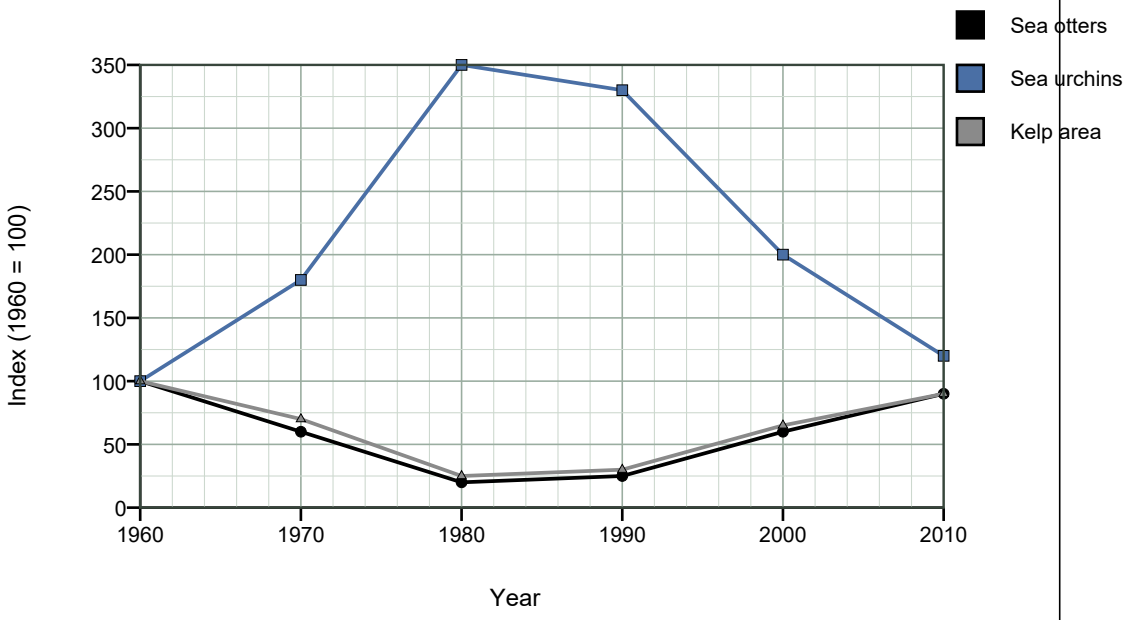
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0 9 Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 9 . 1 Describe the relationship between the three variables shown in Figure 1.

[3 marks]

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0 9 . 2 Explain the relationships shown in Figure 1.

[3 marks]

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0 9 . 3 Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

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2 .....

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0 9 . 4 Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

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1 0 . 1 Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

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1 0 . 2 State what is meant by **biodiversity**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS